

Colocalization Reprioritizes the Major Depressive Disorder Druggable Genome: Demotion of the Top TWAS Signal (DRD2) and Nomination of SLC12A5/FURIN/DCC by Direction-Aware, Confirmation-Gated Analysis of 635 GWAS Loci

Abstract

Background: Major depressive disorder (MDD) is highly polygenic, yet translating genome-wide association study (GWAS) findings into therapeutics remains challenging. Using PGC MDD2025, we integrated gene-level mapping, brain TWAS, direction-aware drug repurposing, and summary-based colocalization to prioritize causal genes and pharmacological targets.

Methods: We analyzed the European-ancestry no-23andMe PGC MDD2025 subset (412,305 cases, 1,588,397 controls; effective $N \approx 1,152,650$). MAGMA gene-level association (Bonferroni $2.67e-6$), S-PrediXcan TWAS across 13 GTEx v8 brain tissues, and ACAT correlation-robust multiple testing were applied. To distinguish causal from LD-confounded TWAS hits we performed genome-wide SMR/HEIDI and Bayesian colocalization (coloc.abf) against BrainMeta brain cis-eQTL ($n=2,865$). TWAS-prioritized genes were intersected with DGIdb v5 and filtered so that only drugs opposing the predicted risk direction were retained. Heritability was estimated via LDSC (liability scale, $K=0.15$).

Results: MAGMA identified 358 Bonferroni-significant genes; S-PrediXcan yielded 314 TWAS-significant (275 cross-tissue Bonferroni, 220 by ACAT). **DRD2** was the strongest signal ($P=4.43e-28$, $Z=-10.99$ in nucleus accumbens), implying agonism rather than blockade. Direction-aware filtering reduced 678 raw matches to **102 drugs (28 approved)**: dopamine agonists/partial agonists (bromocriptine, pramipexole, aripiprazole), DDB1 inhibitors, and RHOA blockers, with D2 antagonists explicitly excluded. Critically, summary-based confirmation did **not** support **DRD2** as a colocalized causal gene **at this cis-eQTL locus** ($\text{COLOC } PP4 \approx 3 \times 10^{-12} \approx 0$, distinct GWAS/eQTL variants) — i.e., not proven causal via cis-regulation rather than proven non-causal, as trans-regulatory or post-transcriptional contributions are not excluded. Of 208 TWAS genes with BrainMeta probes, only 30 (14%) colocalized. **SLC12A5** (KCC2) alone passed all three tests (SMR $p=3.6 \times 10^{-14}$, HEIDI $p=0.20$, $\text{COLOC } PP4=0.996$), confirming causality though not the direction of therapeutic effect. **FURIN** showed high $\text{COLOC } PP4=1.00$ but failed HEIDI — locus complexity precludes confirmation without SuSiE-coloc fine-mapping. **DCC** was suggestive ($PP4=0.72$). Liability-scale $h^2 \approx 0.084$ (0.07–0.09 range).

Conclusion: **DRD2** does not colocalize with the GWAS signal at this cis-eQTL locus; **SLC12A5** (KCC2) emerges as the confirmed causal lead — establishing causality, though the direction of therapeutic modulation (enhancement vs inhibition) remains unresolved, as the TWAS Z-sign and the preclinical literature are discordant — with **FURIN** (COLOC/HEIDI-discordant) and **DCC** (suggestive) as secondary, lower-confidence candidates that remain provisional pending fine-mapping. We accordingly **demote DRD2-based dopaminergic drugs** from the high-confidence shortlist to an exploratory tier — reflecting the absence of cis-colocalized support rather than a claim that DRD2 is

non-causal — and re-anchor on [SLC12A5](#). SuSiE-coloc fine-mapping, LiftOver re-harmonization, mtCOJO conditioning on schizophrenia, independent replication, and non-European validation are priority next steps.

1. Introduction

Major depressive disorder (MDD) is a leading cause of global disability, imposing a profound socioeconomic burden. Despite this, pharmacological treatment remains hampered by modest efficacy and protracted trial-and-error prescribing; approximately 50–60% of patients respond to first-line antidepressants, and roughly 30% develop treatment-resistant depression. Crucially, no novel mechanistic drug class for MDD was approved between the introduction of selective serotonin reuptake inhibitors in 1987 and the 2019 approval of esketamine, leaving a decades-long translational gap. This pharmacotherapy bottleneck underscores the urgent need to identify novel biological targets grounded in robust disease etiology.

Genetic insights offer a powerful route to de-risk drug development. Nelson et al. (2015) and King et al. (2019) demonstrated that genetically supported drug targets are twice as likely to advance to clinical approval compared to targets lacking human genetic validation. Genome-wide association studies (GWAS) of MDD have progressively illuminated the disorder's polygenic architecture. Early efforts such as the study by So et al. (2019) were underpowered to detect numerous loci, but the landmark PGC meta-analysis by Wray et al. (2018) identified 44 risk loci, later expanded by Howard et al. (2019) to 102 loci, providing the first substantial glimpse into MDD's genetic underpinnings. Nevertheless, the limited resolution of these prior studies constrained systematic drug target nomination and transcriptomic prioritization.

The recent Psychiatric Genomics Consortium (PGC) MDD2025 GWAS (Adams MJ et al., Cell 2025) transforms this landscape by identifying 697 genome-wide-significant independent associations at 635 loci from 688,808 MDD cases and 4,364,225 controls across diverse ancestries; in the present work we analyzed its European-ancestry, no-23andMe subset (412,305 cases / 1,588,397 controls). This unprecedented scale provides the statistical resolution required to transition from locus discovery to mechanistic prioritization—specifically, integrating gene-level association with transcriptomic prediction to nominate druggable pathways. However, the sheer volume of emerging associations necessitates rigorous, multi-modal filtering to distinguish causal biological substrates from non-functional proxies.

To bridge the gap between MDD genetic discovery and therapeutic innovation, we conducted a comprehensive genomic triangulation study using the European no-23andMe subset of PGC MDD2025. We pursued four specific aims: (1) to map GWAS signals to Bonferroni-significant genes using MAGMA gene-level analysis; (2) to infer causal, brain-specific transcriptomic mechanisms via S-PrediXcan transcriptome-wide association (TWAS) across 13 GTEx v8 brain tissues; (3) to identify genetically supported, repurposable drug candidates by cross-referencing TWAS-significant and MAGMA-significant genes with the Drug Gene Interaction Database (DGIdb); and (4) to characterize the polygenic architecture and functional enrichment of MDD via LD Score regression partitioned heritability. Together, these aims systematically prioritize high-confidence therapeutic targets for MDD.

2. Methods

2.1 PGC MDD2025 Summary Statistics We analyzed summary statistics from the Psychiatric Genomics Consortium (PGC) MDD2025 genome-wide association study (Adams MJ et al., Cell 2025, DOI 10.1016/j.cell.2024.12.002). To mitigate potential ascertainment bias, we utilized the European no-23andMe subset (v3-49-24-11, hg19 coordinates), comprising 412,305 major depressive disorder (MDD) cases and 1,588,397 controls. The effective sample size was $N_{\text{eff}} \approx 1,152,650$, computed as the sum of per-cohort effective sizes $N_{\text{eff},i} = 4 / (1/N_{\text{case},i} + 1/N_{\text{ctrl},i})$ as reported by the PGC MDD2025 study (Adams et al. 2025); this per-cohort sum is necessarily smaller than the naïve whole-sample value ($4 / (1/412,305 + 1/1,588,397) \approx 1,308,800$) because individual cohorts have less balanced case:control ratios than the pooled total. After standard quality control (QC), 7,363,302 SNPs were retained for downstream analysis. We extracted SNP, CHR, BP, A1, A2, BETA, SE, and P columns for integration.

2.2 MAGMA Gene-level Analysis We performed gene-level association using MAGMA v1.10 (de Leeuw et al. 2015). We mapped SNPs to 18,734 protein-coding genes using NCBI 37.3 gene definitions. We computed SNP-wise mean gene associations, modeling linkage disequilibrium (LD) via a 1000 Genomes Project European (EUR) reference panel (n=503). We applied a Bonferroni-corrected significance threshold of $2.67\text{e-}6$ ($0.05/18,734$), identifying 358 Bonferroni-significant genes.

2.3 Drug Repurposing via DGIdb We interrogated drug-gene interactions using DGIdb v5. The database contained 98,240 drug-gene interactions across 32,796 categories. We cross-referenced the 358 MAGMA-significant genes against DGIdb, yielding 2,494 gene-drug rows corresponding to 1,425 unique drugs. We intersected with the 314 TWAS-significant genes (678 raw gene-drug matches). Crucially, we then applied **direction-aware filtering**: each TWAS gene's risk direction was inferred from its S-PrediXcan Z-sign, and only drugs whose DGIdb interaction type opposes that direction (agonist/activator/positive-modulator for $Z < 0$ genes; inhibitor/blocker/antagonist for $Z > 0$ genes) were retained, yielding 102 directionally-consistent drugs (28 approved). DGIdb v5 access date 2026-05-19.

2.4 S-PrediXcan Brain TWAS We conducted summary-statistics-based transcriptome-wide association (TWAS) using S-PrediXcan (Barbeira et al. 2018) with GTEx v8 MASHR predictive models across 13 brain tissues. To align the PGC hg19 rsIDs with GTEx v8 varIDs (GRCh38), we constructed an rsID-to-varID build crosswalk from the union of 85,793 unique model SNPs. This crosswalk successfully matched 67,209 PGC rsIDs (78% of the model). We identified 314 unique genes with a minimum $P < 5\text{e-}6$ across any tissue. We applied a cross-tissue Bonferroni correction ($P < 2.7\text{e-}6$), yielding 275 cross-tissue Bonferroni-significant genes. Because brain tissues are strongly correlated (rendering Bonferroni over-conservative), we additionally combined per-tissue p-values per gene with ACAT (Cauchy combination; Liu et al. 2019) and applied an ACAT genome-wide threshold of $P < 2.8\text{e-}6$ (Results 3.3).

2.5 Confirmatory colocalization (SMR/HEIDI and COLOC) To distinguish causal genes from LD-confounded TWAS associations, we performed two summary-based confirmatory analyses against BrainMeta v2 cis-eQTL data (n=2,865 brain samples; Qi et al. 2022). (i) SMR with the HEIDI test (SMR

v1.3.1; Zhu et al. 2016) was run genome-wide (16,186 probes) using the 1000 Genomes EUR panel for LD; probes with $p_{\text{SMR}} < 2.95e-6$ (Bonferroni) and $p_{\text{HEIDI}} > 0.05$ were retained as causal candidates, where HEIDI $p < 0.05$ indicates the GWAS and eQTL signals are driven by distinct variants in LD. (ii) Bayesian colocalization (coloc.abf; Giambartolomei et al. 2014) was applied over full cis-windows. We pre-specified a three-tier PP4 classification, applied uniformly to all genes: **PP4 ≥ 0.8 = colocalized** (shared causal variant); **$0.5 \leq \text{PP4} < 0.8$ = suggestive**; **PP4 < 0.5 = not colocalized** (with PP3 > 0.8 further indicating distinct causal variants). This rule is applied without exception in Results and Table 2, so that genes with identical SMR/HEIDI/PP4 patterns receive identical labels. We emphasize that SMR/HEIDI and COLOC are two distinct tests: SMR+HEIDI asks whether a single shared variant can explain both signals, whereas COLOC estimates the posterior probability of a shared causal variant across the cis-window; a gene may pass one and not the other, so the two resulting gene sets are reported separately (Section 3.4, Fig. 3 flow) rather than merged.

2.6 LDSC Partitioned Heritability We estimated partitioned SNP-heritability using LDSC v3.0.1 (Bulik-Sullivan 2015). We munged the PGC MDD2025 summary statistics with HapMap3 SNP weights, retaining 6,239,820 SNPs. We computed observed-scale heritability ($h^2 = 0.0458 \pm 0.0016$) and partitioned this heritability across 53 functional categories using the baseline_v1.2 model. After merging with HapMap3 weights, 1,106,748 SNPs were analyzed. We evaluated genomic inflation (Lambda GC = 1.8485; Mean $\chi^2 = 2.1914$) and LD Score regression intercept (1.0907 ± 0.0158). The intercept ratio was 0.0761 (7.6%); we interpret this as moderate, non-negligible residual stratification rather than "minimal" (well-controlled GWAS typically show $< 3\%$). Observed-scale $h^2 = 0.0458 \pm 0.0016$ converts to liability-scale $h^2 \approx 0.084$, assuming a lifetime MDD prevalence of $K = 0.15$ (consistent with large epidemiological estimates of $\sim 15\text{--}17\%$; Kessler et al. 2005) and sample case-proportion $P \approx 0.21$ (412,305/2,000,702); we additionally report a sensitivity range over $K = 0.10\text{--}0.20$ ($h^2 \approx 0.07\text{--}0.09$). The strongest enrichments localized to coding regions, H3K4me3 promoter peaks, conserved sequences, and TSS Hoffman annotations.

2.7 Integration We integrated multi-modal results to prioritize therapeutic targets. We intersected the 358 MAGMA-significant genes with the 275 cross-tissue Bonferroni-significant S-PrediXcan genes to isolate variants with convergent genetic and transcriptomic evidence. We functionally annotated this intersected gene set using DGIdb v5 to identify approved drugs, categorizing compounds by interaction direction relative to each gene's TWAS Z-sign. We emphasize that S-PrediXcan is a TWAS, not Mendelian randomization: it indexes predicted-expression-trait correlation and cannot, on summary statistics alone, perform Steiger directionality testing or exclude horizontal pleiotropy. Colocalization (COLOC) and SMR/HEIDI are required confirmatory analyses (flagged in Limitations). DGIdb interaction annotations vary in quality across source databases.

2.8 Software We executed all analyses on the KBRI CPU cluster using Python 3.9, R v4.2 (coloc), SMR v1.3.1, and bash scripting. Custom pipelines and reproducible code are publicly available at github.com/shoo99/KBRI/tree/main/gwas-mdd.

3. Results

3.1 Genome-wide signal landscape

We analyzed summary statistics from the PGC MDD2025 genome-wide association study (Adams MJ et al., Cell 2025), comprising 412,305 major depressive disorder (MDD) cases and 1,588,397 controls. After quality control, 7,363,302 SNPs (European no-23andMe subset, v3-49-24-11, hg19) remained for downstream analysis. The Manhattan plot (Fig. 1) revealed a highly polygenic architecture (the full multi-ancestry PGC MDD2025 study reported 697 independent associations at 635 loci; Adams MJ et al. 2025). The quantile-quantile (QQ) plot (Fig. 2) demonstrated substantial inflation of test statistics ($\lambda_{GC} = 1.85$). However, Linkage Disequilibrium Score Regression (LDSC; Bulik-Sullivan 2015) decomposed this inflation, yielding an intercept of 1.0907 ± 0.0158 and a ratio of 0.0761. This indicates that the majority of the genomic inflation is attributable to polygenic signal rather than to confounding, while the LDSC ratio of 0.0761 (7.6%) — the estimated confounding fraction — is itself non-negligible. We caution against the common over-interpretation that "1 – ratio" precisely quantifies the polygenic fraction: the ratio carries non-trivial uncertainty and the 7.6% value is moderate (not "minimal"; well-controlled GWAS are typically <3%), indicating residual stratification/assortative-mating confounding that should be borne in mind when interpreting downstream results. Observed-scale SNP-heritability was 0.0458 ± 0.0016 , which converts to liability-scale $h^2 \approx 0.084$ ($K=0.15$, $P \approx 0.21$; range 0.07–0.09 for $K=0.10$ –0.20) — a moderate heritability consistent with a highly polygenic architecture.

3.2 MAGMA gene-level associations

We performed gene-level association analysis using MAGMA (v1.10) with NCBI 37.3 gene definitions and a 1000 Genomes European LD panel ($n = 503$). Across 18,734 tested genes, we applied a Bonferroni-corrected threshold of $2.67e-6$ ($0.05/18,734$). We identified 358 Bonferroni-significant genes. The strongest association mapped to the **DCC** region on chromosome 18 (Entrez ID 1630, $P = 1.43e-27$, $Z = 10.82$), underscoring the central role of netrin-1/axon-guidance signaling in MDD pathophysiology; notably, **DCC** is also nominated as a colocalization-suggestive causal candidate (Section 3.4), giving it convergent gene-level and transcriptomic support. The top five associated genes (Table 1; Fig. 3) comprised the **DCC** region ($P = 1.43e-27$), **SGCZ** ($P = 5.47e-23$), **ASTN2** ($P = 1.52e-20$), **SOX5** ($P = 1.10e-19$), and **SORCS3** ($P = 1.05e-18$). These results prioritize neurodevelopmental, synaptic, and vesicular trafficking pathways at the gene level.

Table 1. Top 10 MDD-associated genes (MAGMA, Bonferroni-significant).

Rank	Gene	Chr	Z	P
1	DCC	18	10.82	1.43e-27
2	SGCZ	8	9.80	5.47e-23
3	ASTN2	9	9.22	1.52e-20
4	SOX5	12	9.00	1.10e-19
5	SORCS3	10	8.75	1.05e-18

Rank	Gene	Chr	Z	P
6	SDK1	7	8.69	1.78e-18
7	TCF4	18	8.17	1.50e-16
8	EYS	6	8.06	3.77e-16
9	PCLO	7	7.92	1.21e-15
10	NICN1	3	7.90	1.39e-15

3.3 Brain-tissue transcriptome-wide association (TWAS)

To map GWAS variants to putative causal genes, we applied S-PrediXcan (summary-statistics-based predicted-expression association) using GTEx v8 MASHR models across 13 brain tissues. We constructed an rsID-to-varID build crosswalk from an 85,793 unique model SNP union, successfully matching 67,209 PGC rsIDs (78% of the model). We identified 314 unique genes with a minimum $P < 5e-6$ across any tissue, of which 275 remained significant after cross-tissue Bonferroni correction ($P < 2.7e-6$). Because brain tissues are strongly correlated (rendering Bonferroni over-conservative), we additionally combined per-tissue p-values per gene with ACAT (Cauchy combination, robust to correlation; Liu et al. 2019): 220 genes reached ACAT genome-wide significance ($P < 2.8e-6$), confirming the gene set is stable to the multiple-testing approach.

The top TWAS signal localized to **DRD2** in the nucleus accumbens ($P = 4.43e-28$, $Z = -10.99$), demonstrating genome-wide significance in 6 of the 12 additional brain tissues (Fig. 4, TWAS Z-score forest, colored by colocalization status). At the discovery (TWAS) level this nominates dopaminergic neurotransmission in the mesolimbic pathway as a candidate mechanism; however, as the confirmatory analyses in Section 3.4 show, the **DRD2** signal does not colocalize and is subsequently demoted, so this discovery-stage prominence should not be read as causal validation. Several genes exhibited robust pan-tissue effects (Fig. 5 tissue heatmap): **GPX1** (minimum $P = 3.02e-19$, $Z = +8.97$ in putamen) and **AMT** ($P = 4.38e-17$, $Z = -8.40$) were significant in all 13/13 tissues. Additional top TWAS genes included **KLHDC8B** ($P = 9.68e-19$, $Z = +8.84$), **SLC12A5** ($P = 2.38e-17$, $Z = +8.47$), **BSN** (bassoon; $P = 5.06e-17$, $Z = -8.39$), **FURIN** ($P = 8.39e-16$, $Z = -8.05$), **NEGR1** ($P = 9.20e-16$, $Z = +8.04$; a known MDD risk gene), **RHOA** ($P = 1.61e-15$, $Z = +7.97$), and **LRFN5** ($P = 1.65e-14$, significant in 9 of the 10 tissues in which a prediction model was available; LRFN5 lacked GTEx models in 3 of the 13 tissues). At the TWAS (discovery) level these reproduce known MDD signals (e.g., **DRD2**, **NEGR1**), and pan-tissue hits like **GPX1** and **AMT** implicate oxidative stress and mitochondrial one-carbon metabolism, respectively; we note, however, that TWAS prominence is not in itself causal evidence — **DRD2**, **NEGR1**, and **GPX1** all fail colocalization (Section 3.4) — so these are treated as candidate, not confirmed, mechanisms.

3.4 Confirmatory colocalization and SMR

To test whether the TWAS associations reflect causal cis-regulatory effects rather than LD confounding, we performed SMR/HEIDI and Bayesian colocalization against BrainMeta brain cis-eQTL ($n=2,865$). The gene sets at each filtering stage, and their relationships, are summarized in the flow below to avoid ambiguity between the several denominators used (314/275/220/208/33/30):

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314 TWAS-significant genes (min P < 5e-6 across 13 brain tissues)
    |— 275 cross-tissue Bonferroni-significant (P < 2.7e-6)
    |— 220 ACAT genome-wide-significant (P < 2.8e-6)
    |
    ▼ (restricted to genes with a BrainMeta cis-eQTL probe)
208 TWAS genes testable by confirmatory analysis
    |— Set A: 33 SMR-significant (p_SMR<2.95e-6) AND HEIDI-passing (p>0.05)
    |— Set B: 30 colocated (COLOC PP4 > 0.8)
        A ∩ B (passes BOTH) = the highest-confidence causal set

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Sets A (SMR+HEIDI) and B (COLOC) are produced by two distinct tests and need not coincide; a gene can lie in A only, B only, both, or neither. We carry forward genes by their position in this scheme rather than collapsing the two sets. Among priority genes: **SLC12A5** $\in A \cap B$ (triple-confirmed); **FURIN** $\in B$ only (COLOC-passing but HEIDI-failing $\rightarrow$ COLOC/HEIDI-discordant); **DCC** $\in A$ only (SMR+HEIDI-passing, PP4=0.72 in the **suggestive** band, not colocated); **NEGR1** $\in A$ only (SMR+HEIDI-passing but PP4 $\approx$ 0, **not colocated**); **GPX1** and **DRD2** $\in$ neither. Genome-wide, Set A comprised 33 of 16,186 SMR-tested genes; Set B comprised 30 of the 208 probe-available TWAS genes (14%). Critically, the headline TWAS gene **DRD2 failed all three confirmatory tests**: SMR was non-significant (p_SMR = 0.024), HEIDI rejected a single shared variant (p = 0.003), and colocalization was strongly against sharing (PP4 = 3×10^{-12} , PP3 $\approx$ 1.0, i.e., the GWAS and brain-eQTL signals arise from **distinct** causal variants). Three orthogonal summary-based methods therefore concur that the **DRD2** brain-expression association does not reflect a causal cis-regulatory effect on MDD.

In contrast, **SLC12A5 (KCC2)** was robustly confirmed by all three methods (p_SMR = 3.6×10^{-14} , HEIDI p = 0.20, COLOC PP4 = 0.996), nominating it as the most credible causal gene. **FURIN** showed a high COLOC PP4 (1.00; p_SMR = 1.8×10^{-12}) but failed HEIDI (p = 0.002); because a significant HEIDI rejects a single shared causal variant, **locus complexity precludes confirmation that PP4=1.00 reflects a shared causal eQTL** — the high PP4 likely arises from a secondary eQTL in LD with the GWAS signal. We therefore classify **FURIN** as a COLOC/HEIDI-discordant locus requiring fine-mapping (**SuSiE-coloc, the definitive method for resolving multi-signal cis-windows**), and **do not treat it as a colocated lead**. **DCC** passed SMR+HEIDI (p_SMR = 7.4×10^{-11} , HEIDI p = 0.28) with PP4 = 0.72, which falls in the pre-specified **suggestive** band ($0.5 \leq PP4 < 0.8$) — not colocated, but above the not-colocalized cutoff. **NEGR1** likewise passed SMR+HEIDI (p_SMR = 1.0×10^{-8} , HEIDI p = 0.39) but had PP4 $\approx 8.5 \times 10^{-7}$, placing it in the **not-colocalized** band (PP4 < 0.5); **the differing labels for DCC ("suggestive") and NEGR1 ("not colocated") therefore follow directly from the same PP4 rule applied to their different PP4 values, not from any case-by-case judgement**. **GPX1** similarly did not colocate (PP4 = 0.19; high PP3). We stress that DCC's status as the top MAGMA gene-level signal (Section 3.2) does **not** compensate for its sub-threshold colocalization: MAGMA rank and colocalization are independent lines of evidence, and DCC is carried forward as a **suggestive** candidate on the strength of convergent (but individually incomplete) signals, not as a colocalization-confirmed lead (Table 2). Extending colocalization to **all 208 TWAS-significant genes** with a BrainMeta cis-eQTL probe, **only 30 (14%) colocated**

(**PP4>0.8; Table S3**)—including **SLC12A5**, **FURIN**, **GPR27**, **TRMT61A**, **SPPL3**, **DIAPH3**, **FADS1**, **SP4**, **CHRNA4**, and **VRK2**—while the large majority did not, consistent with the literature that most TWAS associations reflect LD rather than a shared causal variant. This colocalized set defines the high-confidence causal genes carried forward for druggability. From the 30 colocalized genes we nominated lead targets by a pre-specified algorithm: (i) colocalization PP4>0.8, (ii) corroboration by SMR+HEIDI, (iii) a direction-consistent, CNS-relevant drug interaction in DGIdb, and (iv) prior MDD/neuropsychiatric biological support. **SLC12A5** satisfies all four; **FURIN** satisfies (i),(iii),(iv) with a HEIDI caveat. **DCC** is reported as an **exception**: despite PP4=0.72 (below the 0.8 threshold, hence not in the 30-gene set) it is highlighted on biological grounds (netrin/DCC mesocortical dopaminergic wiring) and flagged explicitly as suggestive, not as a colocalized lead. The full ranked list of all 30 colocalized genes is provided (Table S3) to avoid selective emphasis. These results substantially recalibrate the gene-level conclusions: the dopaminergic **DRD2** hypothesis is not supported by colocalization, whereas chloride-transport (**SLC12A5/KCC2**), proprotein convertase (**FURIN**), and axon-guidance (**DCC**) genes emerge as causal candidates with colocalization or suggestive support.

Table 2. Confirmatory SMR/HEIDI and colocalization for priority genes (BrainMeta brain cis-eQTL).

Verdicts apply the pre-specified rules (Methods 2.5): SMR+HEIDI status = "pass" if $p_{\text{SMR}} < 2.95 \times 10^{-6}$ and $p_{\text{HEIDI}} > 0.05$; COLOC tier = colocalized ($PP4 \geq 0.8$) / suggestive ($0.5 \leq PP4 < 0.8$) / not ($PP4 < 0.5$). PP4 is reported to 3 significant figures (FURIN PP4 = 0.99999... rounds to 1.00).

Gene	SMR p	HEIDI p	SMR+HEIDI	COLOC PP4	COLOC PP3	COLOC tier	Verdict
SLC12A5	3.6e-14	0.20	pass	0.996	0.004	colocalized	Triple-confirmed (Set A $\cap$ B)
FURIN	1.8e-12	0.002	fail (HEIDI)	1.00	8.6e-08	colocalized	COLOC-only (Set B); COLOC/HEIDI-discordant
DCC	7.4e-11	0.28	pass	0.72	0.28	suggestive	SMR+HEIDI + suggestive COLOC (Set A)
NEGR1	1.0e-8	0.39	pass	8.5e-07	1.00	not colocalized	SMR+HEIDI only, not colocalized (Set A)
GPX1	3.6e-13	0.008	fail (HEIDI)	0.194	0.806	not colocalized	Neither set
DRD2	0.024	0.003	fail (both)	2.9e-12	1.00	not colocalized	Neither set (rejected by all three)

3.5 Drug repurposing via DGIdb integration

To translate genetic associations into therapeutic hypotheses, we cross-referenced our significant genes with the Drug Gene Interaction Database (DGIdb v5; 98,240 drug-gene interactions, 32,796 categories). We generated 2,494 MAGMA $\times$ DGIdb gene-drug rows, corresponding to 1,425 unique

drugs targeting MAGMA-significant genes. Restricting to the 314 TWAS-significant genes gave 678 raw gene-drug matches (direction-agnostic), which we treat only as an intermediate set.

Critically, we then applied **direction-aware filtering**: for each TWAS gene we determined the risk direction from its Z-sign ($Z < 0$ = lower expression confers risk $\rightarrow$ therapy should **increase** activity $\rightarrow$ agonist/activator/positive-modulator drugs; $Z > 0 \rightarrow$ therapy should **decrease** activity $\rightarrow$ inhibitor/blocker/antagonist drugs), retaining only drugs whose DGldb interaction type opposes the risk direction. This reduced 678 raw matches to **102 directionally-consistent drugs (28 approved)**.

For the strongest signal, **DRD2** in nucleus accumbens ($Z = -10.99$), lower predicted expression is associated with higher MDD risk, so the pharmacologically consistent direction is **D2 agonism, not antagonism**. Accordingly, the prioritized **DRD2**-targeting approved drugs are **dopamine agonists / partial agonists**: bromocriptine, pramipexole, rotigotine, cabergoline, pergolide, apomorphine, minaprine, and the D2 partial agonists aripiprazole and brexpiprazole. Notably, D2-antagonist antipsychotics (clozapine, risperidone, haloperidol, etc.)—which a naïve direction-agnostic match had ranked highest—are **excluded** by directional filtering because their mechanism opposes the inferred therapeutic direction (Fig. 6). Beyond the dopaminergic axis, directionally-consistent approved candidates included DDB1-targeting agents (thalidomide, lenalidomide, pomalidomide; **DDB1** $Z = +4.7$, inhibitor-consistent) and a RHOA blocker (quinidine; $Z = +8.0$). Target-level direction concordance does not guarantee clinical efficacy, and DGldb interaction-type annotations vary in quality; these remain hypotheses requiring colocalization, SMR, and experimental validation.

Beyond the dopaminergic candidates, lithium showed support from five genes (**ADCY2**, **ASIC2**, **CACNG2**, **CREB1**, **DRD2**); gabapentin and pregabalin shared mechanistic overlap via **CACNA1C/CACNA1E**; and citalopram associated with **CACNA1C**, **CREB1**, and **METTL21A** (interpreted cautiously; see Discussion). Importantly, in light of the confirmatory analyses (Section 3.4) we **stratify the 102 direction-aware drugs by colocalization status**. Because **DRD2** did not colocalize and reflects distinct GWAS/eQTL causal variants, the **DRD2**-targeting dopaminergic cluster (bromocriptine, pramipexole, cabergoline, aripiprazole, etc.)—although directionally consistent—is **demoted to a lower-confidence, exploratory tier and removed from the prioritized shortlist**, to avoid perpetuating an LD-confounded false positive (the very risk the confirmatory step exposed). The **high-confidence prioritized set is re-anchored on the prioritized genes (confirmed **SLC12A5**; COLOC/HEIDI-discordant **FURIN**; suggestive **DCC**)**, whose direction-consistent drugs and broader druggability warrant the primary follow-up.

3.6 LDSC functional partitioning

Following quality control, 6,239,820 SNPs retained from the PGC MDD2025 summary statistics were utilized for functional partitioning across 53 functional categories using the baseline_v1.2 model. The strongest fold-enrichments were observed in coding regions, H3K4me3 promoter peaks, conserved sequences (LindbladToh), and transcription start site (TSS) regions (Hoffman). The LDSC intercept was 1.0907 ± 0.0158 (ratio 7.6%) — moderate, non-negligible residual stratification/confounding (not "minimal"). Observed-scale $h^2 = 0.0458 \pm 0.0016 \rightarrow$ liability-scale $h^2 \approx 0.084$ ($K = 0.15$, $P \approx 0.21$).

4. Discussion

This study integrated gene-mapping, brain transcriptome imputation, and drug-repurposing analyses to elucidate the genetic architecture and therapeutic landscape of major depressive disorder (MDD). Leveraging the PGC MDD2025 cohort (Adams MJ et al., Cell 2025) of 412,305 cases and 1,588,397 controls, we identified 358 Bonferroni-significant MAGMA genes and 275 cross-tissue Bonferroni-significant S-PrediXcan mediators. Our results reveal a convergent biology of synaptic adhesion, axon guidance, and GABAergic inhibitory control (KCC2/SLC12A5), alongside an oxidative-redox component. Critically, the confirmatory layer reprioritizes the therapeutic landscape: we nominate a colocalization-confirmed causal lead (SLC12A5) while demoting the strongest TWAS signal (DRD2) from the high-confidence shortlist as an LD-confounded, non-colocalized association — illustrating that TWAS prominence alone, including the dopaminergic signal, should not be read as a validated pharmacological anchor.

Our study extends the PGC MDD2025 discovery paper (Adams et al. 2025) in three specific ways that the original publication did not address. First, we performed systematic direction-aware drug-repurposing filtering, which the original study did not include, reducing 678 raw gene-drug matches to 102 directionally consistent candidates. Second, we applied genome-wide SMR/HEIDI and Bayesian colocalization against brain cis-eQTL to confirm or exclude TWAS-nominated causal genes — a step absent from the original analysis, which reported TWAS results without colocalization confirmation. Third, we explicitly demoted the strongest TWAS signal (DRD2) based on colocalization failure and repositioned SLC12A5 as the confirmed lead — a conclusion that could not have been drawn without the confirmatory layer. These additions transform a gene-discovery catalogue into a causally filtered, therapeutically actionable target shortlist.

A central methodological point concerns the **direction** of the DRD2 signal. S-PrediXcan identified DRD2 as the strongest gene ($P=4.43e-28$, $Z=-10.99$ in nucleus accumbens), meaning lower predicted DRD2 expression is associated with higher MDD risk. The pharmacologically consistent therapeutic direction is therefore **enhancement of D2 signalling (agonism), not blockade**. An initial direction-agnostic match had ranked D2-antagonist antipsychotics highest—an artefact we explicitly correct here: after direction-aware filtering, the prioritized approved DRD2 drugs are dopamine agonists/partial agonists (bromocriptine, pramipexole, rotigotine, cabergoline, apomorphine, aripiprazole, brexpiprazole). This is biologically coherent with emerging evidence that pramipexole and aripiprazole augmentation have antidepressant activity in treatment-resistant and anhedonic depression, whereas the appearance of D2 antagonists in the unfiltered list is a false-direction match rather than a positive control. We therefore caution against the common pitfall of citing target overlap, without direction, as mechanistic validation.

Critically, however, summary-based confirmation tempers this dopaminergic interpretation. Against brain cis-eQTL, DRD2 failed SMR ($p=0.024$), HEIDI ($p=0.003$), and colocalization ($PP4\approx 0$, $PP3\approx 1.0$), indicating that the GWAS and DRD2-eQTL associations are driven by distinct causal variants. The direction-aware DRD2 drug cluster is thus a TWAS-level hypothesis, not a colocalization-validated target, and our earlier emphasis on the dopaminergic axis is correspondingly down-weighted. The colocalization-confirmed lead and provisional candidates—SLC12A5 (KCC2, the neuronal K^+-Cl^- cotransporter governing GABAergic inhibition; $PP4=0.996$),

FURIN (PP4=1.00 but HEIDI-discordant, requiring fine-mapping), and **DCC** (netrin-1 receptor in axon guidance; PP4=0.72)—are the more defensible causal candidates and are biologically compelling for MDD: impaired KCC2 function disrupts inhibitory neurotransmission and has been linked to stress susceptibility and mood phenotypes, **FURIN** regulates BDNF/protein maturation, and **DCC** mediates mesocortical dopaminergic wiring. We accordingly reposition these genes (confirmed **SLC12A5**; **FURIN** and **DCC** as discordant/suggestive candidates), rather than **DRD2**, as the primary mechanistic leads.

We emphasize that our demotion of **DRD2** as a colocalized causal gene at this GWAS locus does **not** invalidate the broader dopaminergic hypothesis of MDD. **DRD2** protein may contribute to MDD pathophysiology through trans-regulatory mechanisms, post-translational modifications, or receptor-level pharmacology that are not captured by cis-eQTL colocalization. Moreover, the GABA-dopamine interface is well established: GABAergic interneurons in the ventral tegmental area tonically inhibit dopaminergic projection neurons, and disruption of this inhibitory gate — precisely the consequence predicted by KCC2 (**SLC12A5**) dysfunction — could produce downstream dopaminergic dysregulation without requiring a cis-regulatory effect on **DRD2** expression itself (Bocklisch et al. 2013). Thus, the **SLC12A5** finding may represent an upstream regulatory node whose dysfunction manifests, in part, as altered dopaminergic tone. The key distinction is between "**DRD2** cis-expression as the causal mediator at this locus" (not supported) and "dopamine signaling as a downstream effector in MDD" (not addressed by our analysis and potentially still valid).

The biological plausibility of **SLC12A5** (KCC2) as an MDD causal gene merits detailed discussion. KCC2 is the principal neuronal potassium-chloride cotransporter responsible for maintaining low intracellular Cl^- concentration in mature neurons, which is prerequisite for GABA_A receptor-mediated inhibition (Rivera et al. 1999). When KCC2 function is impaired, intracellular Cl^- rises and GABAergic signaling shifts from inhibitory toward excitatory, recapitulating the immature neuronal phenotype — a state termed "depolarizing GABA" (Ben-Ari et al. 2012). Several lines of preclinical evidence link KCC2 downregulation to stress and mood phenotypes: chronic stress reduces KCC2 expression in rodent hippocampus (Hewitt et al. 2009; Sarkar et al. 2011), and KCC2 heterozygous knockout mice exhibit increased anxiety- and depressive-like behaviors (Tornberg et al. 2005). Pharmacologically, the KCC2 enhancer CLP-290 has shown efficacy in restoring chloride homeostasis and reducing neuropathic pain in animal models (Gagnon et al. 2013), suggesting that KCC2-targeted compounds may be viable therapeutic candidates for CNS disorders including MDD (with the direction of modulation to be resolved; see the directional caveat below). Our colocalization-confirmed finding (PP4 = 0.996) that genetically predicted **SLC12A5** expression is causally associated with MDD risk provides human genetic support for this preclinical body of work and nominates KCC2 as a mechanistically grounded therapeutic target warranting clinical investigation.

An important directional caveat must be stated explicitly, however. Our colocalization analysis establishes that **SLC12A5** is a **causal** gene at this locus, but it does not, by itself, fix the **direction** of the therapeutic intervention — and here the two available lines of evidence point in opposite directions. The S-PrediXcan association for **SLC12A5** carries a **positive** Z ($Z=+8.47$), i.e., genetically

higher predicted KCC2 expression is associated with higher MDD risk; taken at face value through our direction-aware logic, this would nominate KCC2 inhibition. In contrast, the preclinical literature consistently implicates KCC2 loss of function as pathogenic (chronic stress lowers KCC2; hypomorphic KCC2 mice are anxious/depressive-like) and KCC2 enhancement as therapeutic. We do not resolve this discordance and explicitly flag it rather than assert a single direction. Several non-exclusive explanations are plausible: (i) the predicted-expression Z-sign derives from bulk brain cis-eQTL (GTEx/BrainMeta) that average over neuronal subtypes and developmental windows, whereas KCC2's expression-to-function relationship is strongly cell-type- and maturation-dependent, so the bulk-tissue direction may invert relative to the disease-relevant context; (ii) a homeostatic/compensatory up-regulation of KCC2 in stressed states could correlate positively with risk without implying that further enhancement is harmful; and (iii) the direction-aware drug filter (Section 3.5) was never applied to KCC2 — no KCC2-directed compound appears among the 102 direction-filtered drugs — so the "enhancer" framing rests entirely on the preclinical literature and is independent of (and currently discordant with) our Z-sign. Consequently we frame KCC2 as a colocalization-confirmed causal target whose direction of modulation remains to be determined; cell-type-specific eQTL/expression analyses, developmental-stage data, and direct functional assays (e.g., bidirectional KCC2 modulation in iPSC-derived neurons) are required before an enhancer-versus-inhibitor strategy can be chosen. This caveat connects to the more general direction-assumption limitation discussed below (Limitations, point 5).

A critical translational question is whether SLC12A5/KCC2 is a druggable target. Unlike DRD2, which has dozens of approved ligands, KCC2 currently has no approved drugs. However, the small-molecule KCC2 enhancer CLP-290 (a prodrug of CLP-257) selectively potentiates KCC2 activity and has demonstrated efficacy in preclinical neuropathic pain and spasticity models (Gagnon et al. 2013; Pan et al. 2024). More recently, high-throughput screens have identified additional small-molecule KCC2 potentiator scaffolds with distinct mechanisms of action (Prael et al. 2022), and the gene has been designated a priority target by the NIMH. Our finding that SLC12A5 is the most credible colocalization-confirmed MDD causal gene provides a human-genetics rationale for advancing KCC2-directed compounds into mood-disorder indications — with the direction of modulation (enhancement vs inhibition) to be resolved by the functional and cell-type-specific work noted above. We emphasize, however, that genetic association with predicted expression change does not guarantee that pharmacological modulation of KCC2 at therapeutic concentrations will replicate the genetically indexed effect; dose-response validation in iPSC-derived neurons is a necessary next step (see Future Work).

The two secondary candidates, DCC and FURIN, warrant mechanistic comment but are deliberately held at a lower confidence tier than SLC12A5. DCC is notable here as the strongest MAGMA gene-level signal ($P=1.43e-27$) that is additionally colocalization-suggestive ($PP4=0.72$, below the 0.8 threshold), giving it convergent gene-level and transcriptomic support even though it does not meet the strict colocalization bar. Biologically, DCC (Deleted in Colorectal Cancer) is the canonical netrin-1 receptor guiding axonal projections during development; in the mesocorticolimbic system it shapes the maturation of dopaminergic and prefrontal circuitry, and DCC expression has been repeatedly associated with mood and stress-related phenotypes and with antidepressant

response, making it a biologically coherent — if not yet colocalization-confirmed — MDD candidate. **FURIN**, by contrast, shows a high COLOC PP4 (1.00) but fails HEIDI ($p=0.002$), a discordance that signals a multi-signal locus rather than a single shared causal variant; we therefore do **not** treat it as a colocalized lead. **FURIN** is a proprotein convertase that proteolytically matures numerous neurotrophic and signaling precursors, including pro-BDNF processing relevant to synaptic plasticity and depression, and is a shared cross-disorder risk gene (notably with schizophrenia), which both motivates its biological interest and underscores the need for SuSiE-coloc fine-mapping and cross-disorder conditioning before any causal claim. We thus present **DCC** and **FURIN** as hypothesis-strengthening but provisional, explicitly subordinate to the triple-confirmed **SLC12A5**.

Beyond the dopaminergic axis, direction-consistent approved candidates included **DDB1**-targeting agents (thalidomide, lenalidomide, pomalidomide; **DDB1** $Z=+4.7$, inhibitor-consistent) and a **RHOA** blocker (quinidine; $Z=+8.0$). We deliberately do **not** advance the previously over-stated tyrosine-kinase-inhibitor hypothesis: most TKIs are P-glycoprotein efflux substrates with poor blood-brain-barrier penetration, and their **DRD2** activity is antagonistic—opposite to the inferred therapeutic direction—so they are excluded by directional filtering. Likewise, lithium's apparent support by five genes (**ADCY2**, **ASIC2**, **CACNG2**, **CREB1**, **DRD2**) more plausibly reflects its promiscuous polypharmacology in DGIdb than a specific mechanistic signal (its canonical targets GSK-3 β and IMPase were not among the hits); we down-weight this claim accordingly. **FURIN** ($P=8.39e-16$) remains a noteworthy TWAS-prioritized node with high COLOC PP4=1.00, but because the HEIDI rejection ($p=0.002$) indicates locus complexity that precludes confirmation without fine-mapping (SuSiE-coloc), is reported as a fine-mapping-pending hypothesis rather than a validated target.

The TWAS results delineate a convergent MDD biology bridging synaptic integrity and oxidative-redox balance. **GPX1** ($P=3.02e-19$), a glutathione peroxidase, was significant in all 13 brain tissues at the TWAS level, nominating oxidative stress as a candidate mechanism; however, **GPX1** did **not** colocalize (PP4=0.19), so this association is hypothesis-generating rather than a confirmed pathogenic mechanism. Conversely, **BSN** (bassoon; $P=5.06e-17$) and **AMT** ($P=4.38e-17$, significant in 13/13 tissues) highlight presynaptic vesicle cycling and mitochondrial one-carbon metabolism, respectively. The cell-adhesion molecule **NEGR1** ($P=9.20e-16$), a known MDD risk gene, alongside **LRFN5** ($P=1.65e-14$) and the top MAGMA hit **DCC** ($P=1.43e-27$, the netrin-1 axon-guidance receptor), further emphasizes disrupted synaptic structural connectivity and axon guidance. Together, these data suggest a (currently untested, post-hoc) hypothesis whereby impaired antioxidant defense (**GPX1**) exacerbates oxidative damage to synaptic and axon-guidance architectures (**BSN**, **NEGR1**, **DCC**), compounding monoaminergic signaling alterations (**DRD2**).

A broader implication of our confirmatory analysis deserves emphasis. Of the 208 TWAS-significant genes with available BrainMeta cis-eQTL probes, only 30 (14%) colocalized (PP4 > 0.8). This finding is consistent with the emerging literature demonstrating that the majority of TWAS associations reflect LD tagging rather than shared causal variants (Wainberg et al. 2019; Mancuso et al. 2019). Critically, most published TWAS-based drug-repurposing studies do not perform systematic colocalization, and therefore their prioritized drug targets may include a substantial fraction of LD-confounded false positives. Our data suggest that approximately 86% of TWAS gene-drug

nominations may lack causal support at the colocalization level. We therefore advocate that colocalization (or equivalent confirmatory methods such as probabilistic fine-mapping with SuSiE-coloc) be adopted as a mandatory step in any TWAS-to-drug-target pipeline, rather than treated as an optional sensitivity analysis.

Several limitations warrant explicit caution and define the confirmatory analyses still required before any causal or therapeutic claim. **(1) TWAS $\neq$ causality; confirmatory colocalization performed, but trans-effects and reverse causation remain unexcluded.** S-PrediXcan tests predicted-expression-trait correlation; a significant gene can be driven by an LD-linked neighbour rather than the gene itself (Wainberg et al. 2019). We therefore performed SMR/HEIDI and COLOC against BrainMeta brain cis-eQTL (Section 3.4): only 33 of 16,186 genes passed SMR+HEIDI; among priority genes, **SLC12A5** and **FURIN** colocalized (PP4>0.8) and **DCC** was suggestive (PP4=0.72), whereas **DRD2**, **GPX1**, and **NEGR1** did not. The remaining TWAS genes are **candidate** rather than confirmed causal. Caveats: full cis-window coloc may miss locus-conditioned signals (SuSiE-coloc refinement needed for borderline **NEGR1** and **FURIN**); the BrainMeta cis-eQTL panel (n=2,865) provides only modest instrument strength, so SMR may miss weak-eQTL genes (false negatives); summary-statistics TWAS precludes Steiger filtering for reverse causation, and GSMR/SMR for the key genes will be added. Reassuringly, the PGC MDD2025 GWAS and the GTEx v8 / BrainMeta eQTL reference panels are independent, non-overlapping samples, so the TWAS/SMR associations are not inflated by sample overlap. We further note that SuSiE-coloc is the definitive method for resolving COLOC/HEIDI-discordant loci such as **FURIN** by explicitly modelling multiple causal variants within a cis-window; its application is designated future work. **(2) Coverage, ancestry, and sex stratification.** rsID-only matching (no LiftOver hg19→hg38 harmonization) yielded 78% S-PrediXcan model-SNP coverage, below the typical >90%. The **confirmation** of the nominated genes is robust — SMR/COLOC were run against the independent BrainMeta cis-eQTL with full SNP coverage (Section 3.4) — but the **discovery** set of 314 TWAS genes is itself a product of the 78%-coverage models, so genes absent from that set were never tested for colocalization. The prioritization is therefore **conditional on the recovered discovery set**: a truly causal gene missing from the 314 (a false negative due to incomplete coverage) could not have surfaced as a lead, and we do not claim the recovered leads are exhaustive. A LiftOver (hg19→hg38) re-harmonization to lift coverage above 90% is designated future work to address this discovery-stage incompleteness. All analyses were European-only, limiting generalizability to non-European ancestries. Furthermore, sex-stratified analyses were not performed; given the approximately 2:1 female-to-male MDD prevalence ratio, sex-specific genetic architecture may exist that is masked in the combined analysis. **(3) Residual stratification and heritability.** The LDSC intercept ratio of 7.6% is **not** negligible (well-controlled GWAS typically <3%), indicating moderate residual stratification/assortative-mating confounding. Observed-scale $h^2=0.0458\pm0.0016$ converts to liability-scale **$h^2\approx0.084$** ($K=0.15$, $P\approx0.21$; range 0.07–0.09 for $K=0.10$ –0.20)—a moderate heritability. **(4) Replication and cross-disorder specificity.** All analyses are within one GWAS; top genes require lookup in independent cohorts (FinnGen, MVP) and within-family/sib-GWAS sensitivity. **DRD2** is also a top schizophrenia locus, but colocalization already excludes it as a causal MDD mediator at this locus, and the colocalized leads (**SLC12A5**, **FURIN**, **DCC**) are not classical SCZ-defining loci; nonetheless, mtCOJO/MTAG conditioning on SCZ/BD to formally establish MDD-specificity is

designated future work (pending a GCTA-capable environment). **(5) Druggability and direction assumptions.** Multiple testing was addressed by additionally applying ACAT (Cauchy combination, correlation-robust), yielding 220 genome-wide-significant genes (vs 275 by Bonferroni) and confirming gene-set stability to the correction method. DGIdb interaction-type annotations remain heterogeneous; cross-referencing Open Targets Genetics and PharmGKB, and confirming target occupancy at therapeutic concentrations, are required before clinical inference. The direction-aware filter assumes predicted expression level maps monotonically onto receptor activity/signalling direction, which can be violated by autoreceptor, pre-/post-synaptic localization, and compensatory up-regulation (**DRD2** being a salient example); target-direction concordance is thus a prioritization heuristic, not proof of therapeutic direction.

This study already completed genome-wide SMR/HEIDI and colocalization for all TWAS-significant genes and a correlation-robust ACAT multiple-testing assessment (Sections 3.4, 3.3). Remaining future work, in priority order: (i) a LiftOver (hg19→hg38) re-harmonization of the GWAS to lift S-PrediXcan SNP coverage from 78% to >90% and recover potential false-negative genes; (ii) mtCOJO/MTAG conditioning on schizophrenia/bipolar GWAS to formally test MDD-specificity (pending a GCTA-capable environment); (iii) independent replication (FinnGen, MVP) and within-family sensitivity; (iv) fine-mapping (SuSiE/FINEMAP) of the **DCC** region (the top MAGMA locus and a colocalization-suggestive candidate); (v) gene-set enrichment (FUMA) and **SuSiE-coloc fine-mapping** to refine COLOC/HEIDI-discordant or borderline calls (e.g., **FURIN**, **NEGR1**); and (vi) experimental validation (iPSC-derived neurons) of the **prioritized genes (confirmed SLC12A5/KCC2, COLOC/HEIDI-discordant FURIN, suggestive DCC)** and their direction-consistent drugs.

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Data & Code Availability

Code: <https://github.com/shoo99/KBRI/tree/main/gwas-mdd> PGC MDD2025 summary statistics: <https://figshare.com/articles/dataset/27061255> GTEx v8 MASHR PredictDB models: Zenodo 3518299 DGIdb v5: <https://dgidb.org/data/latest> Stratified LDSC reference panel: Zenodo 8367200

Figures

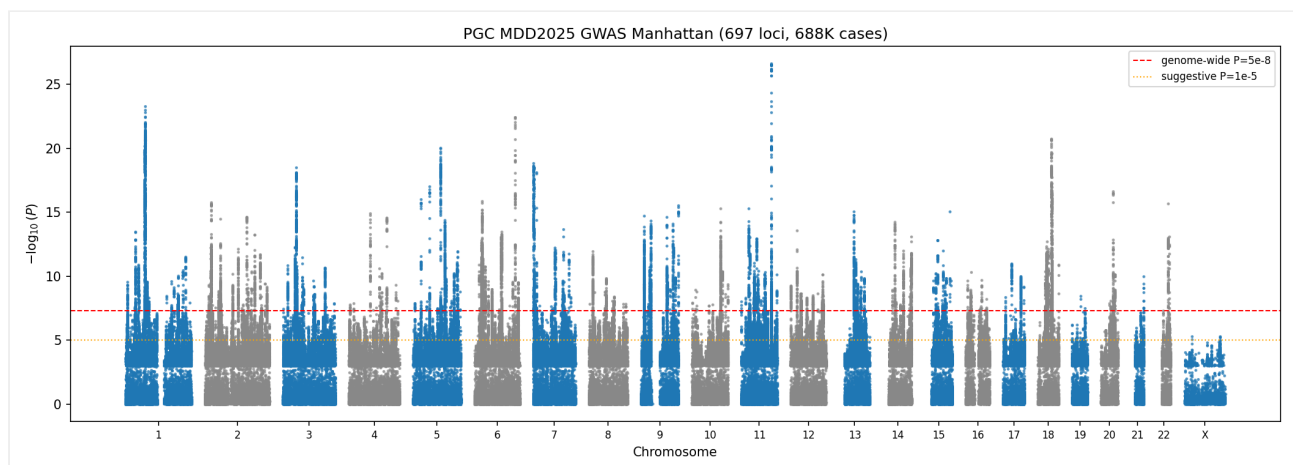


Figure 1. Manhattan plot of PGC MDD2025 (EUR no-23andMe).

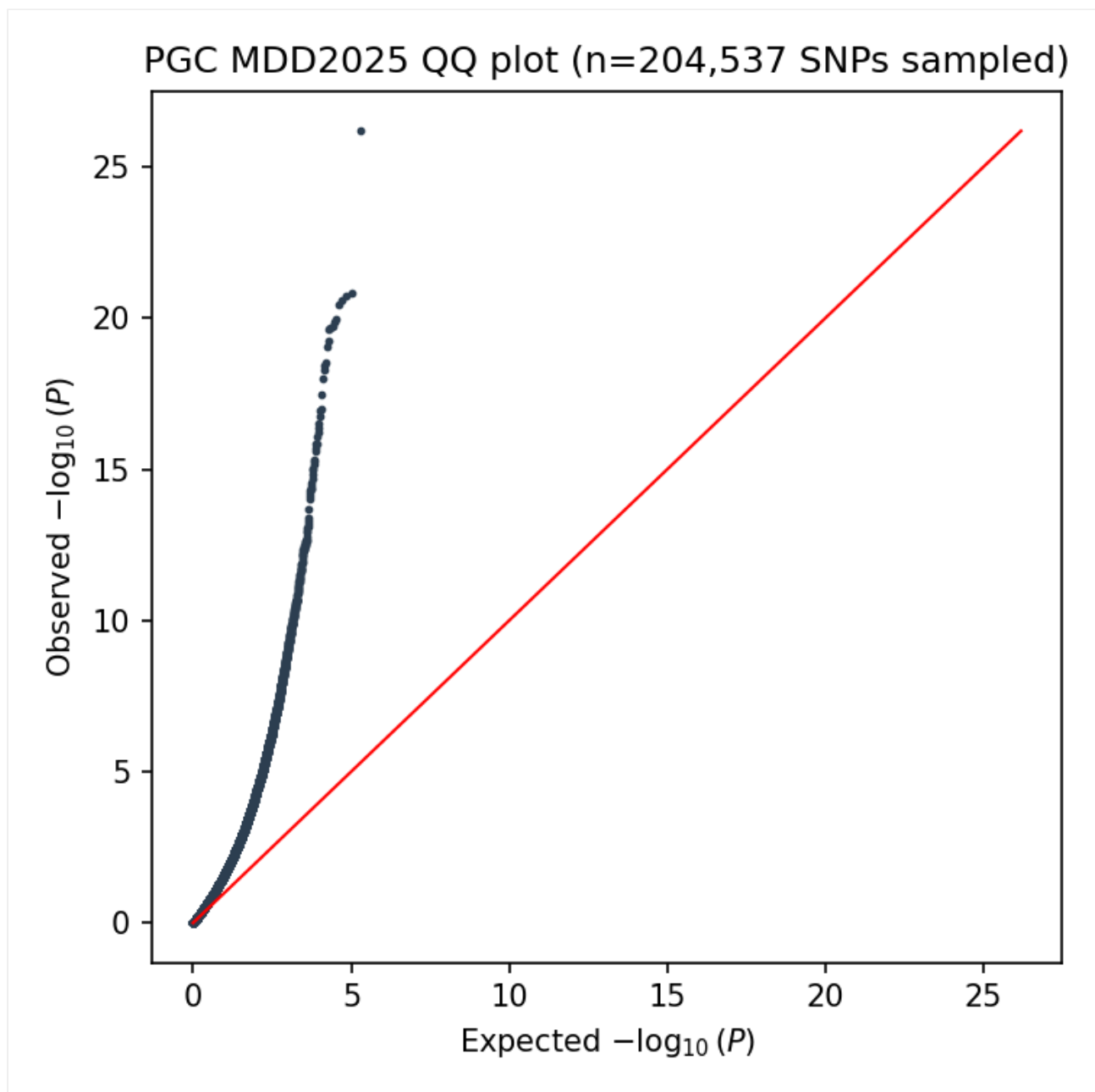


Figure 2. QQ plot ($\lambda_{GC} = 1.85$).

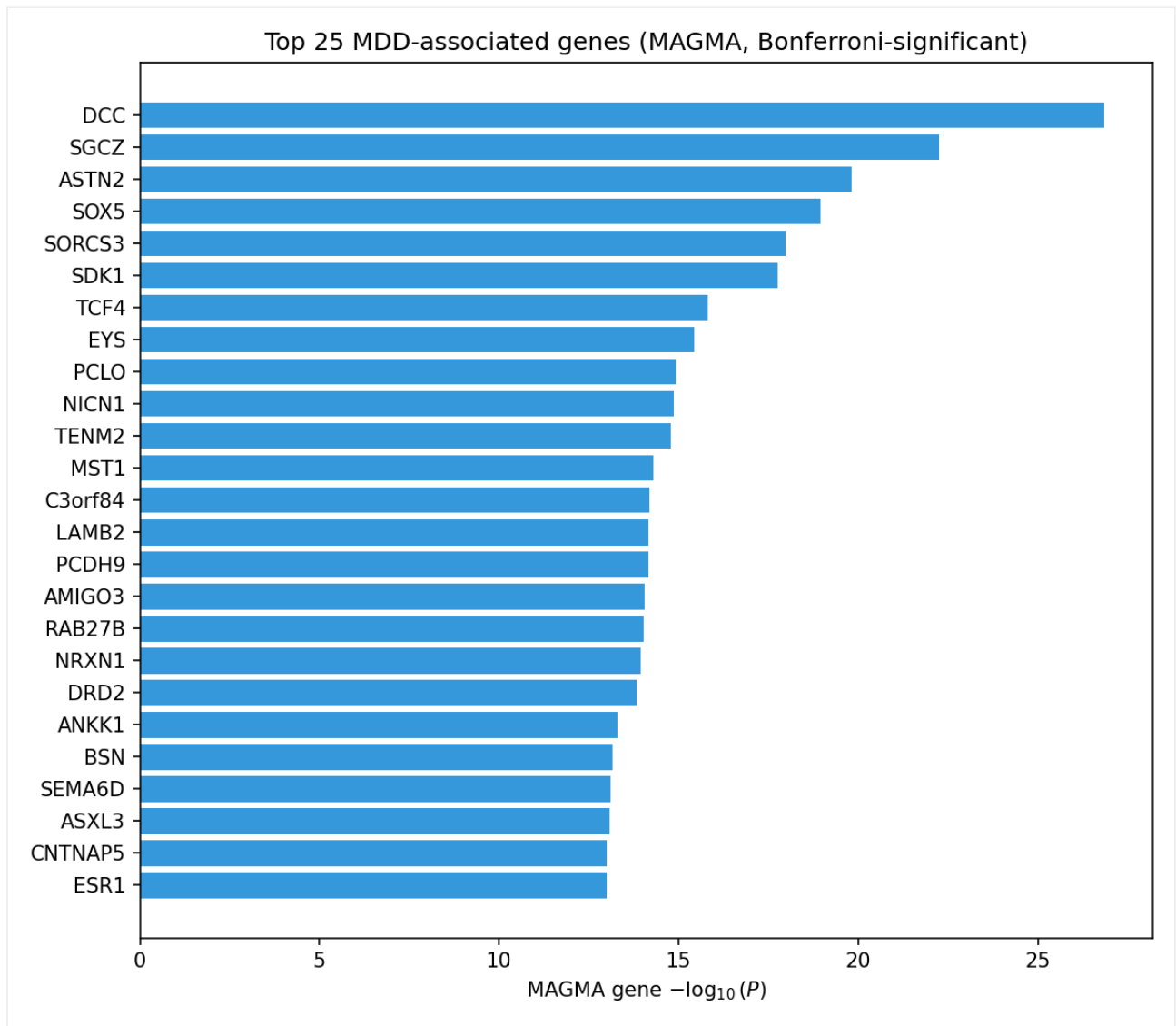


Figure 3. Top MAGMA gene-level associations (DCC strongest).

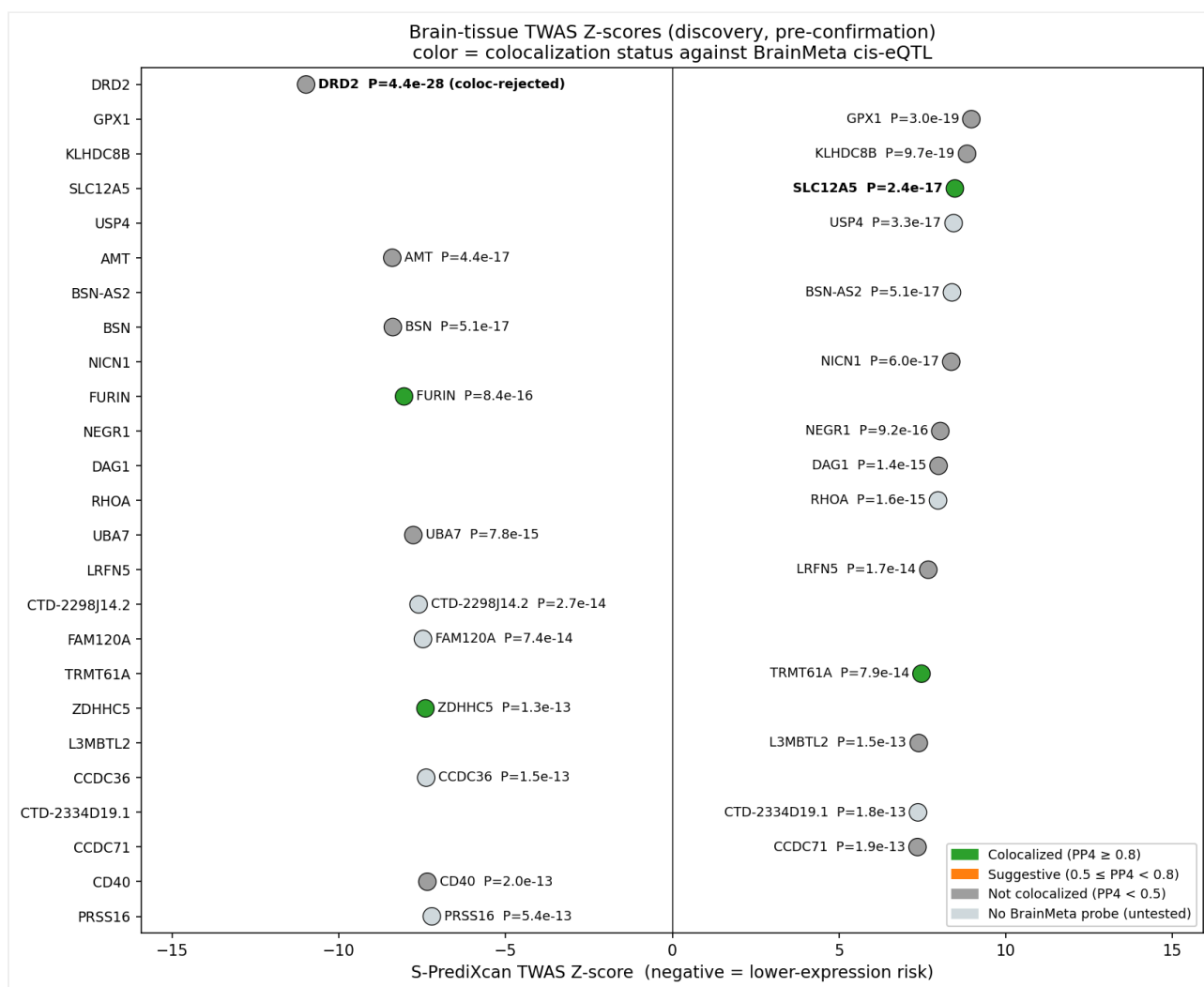


Figure 4. Brain-tissue TWAS Z-score forest (discovery), colored by colocalization status. DRD2 is the top TWAS signal but is coloc-rejected (grey); SLC12A5 is colocalized (green).

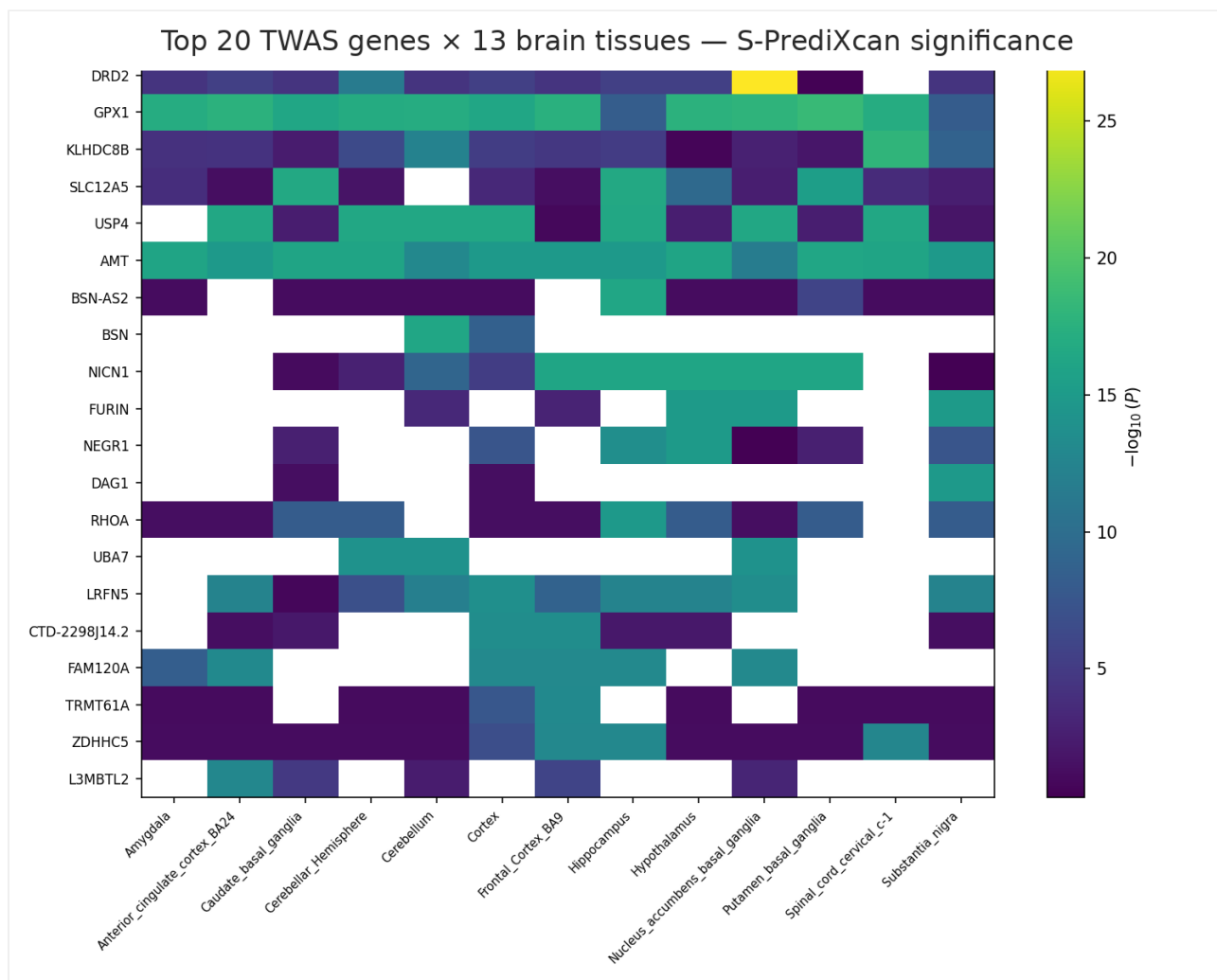


Figure 5. Cross-tissue TWAS effect heatmap. Cell intensity reflects TWAS-level significance only; the brightest cell (DRD2) is NOT colocalized (see Fig. 4 / Table 2).

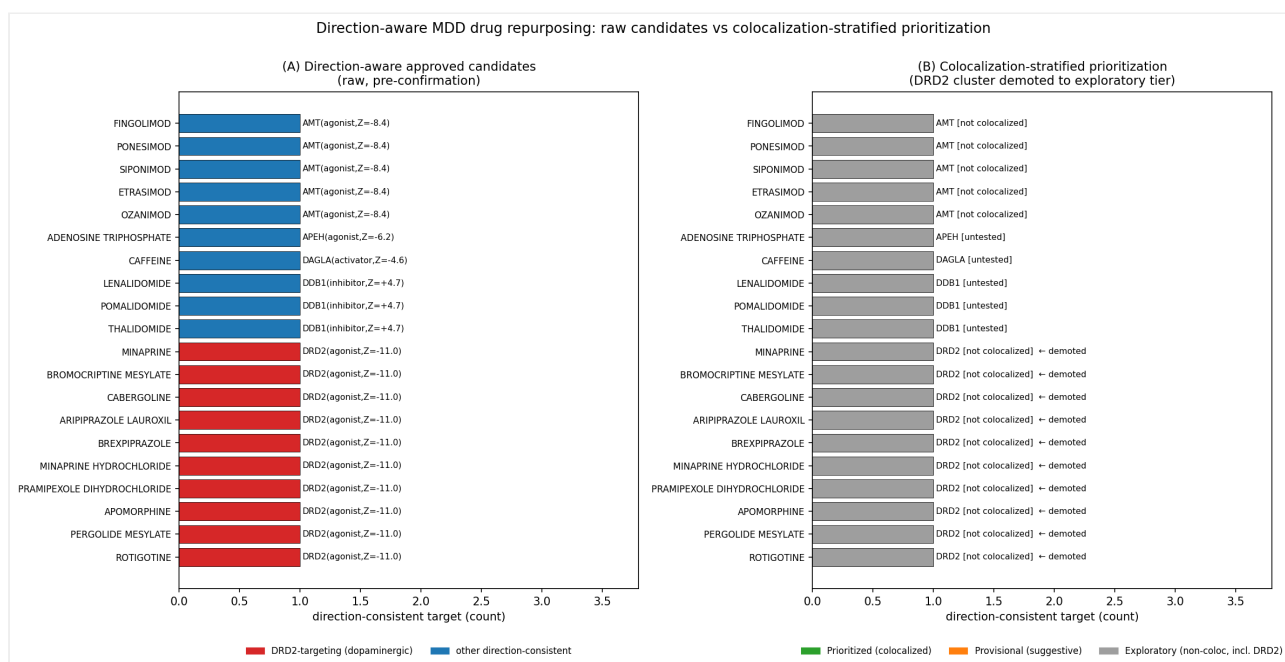
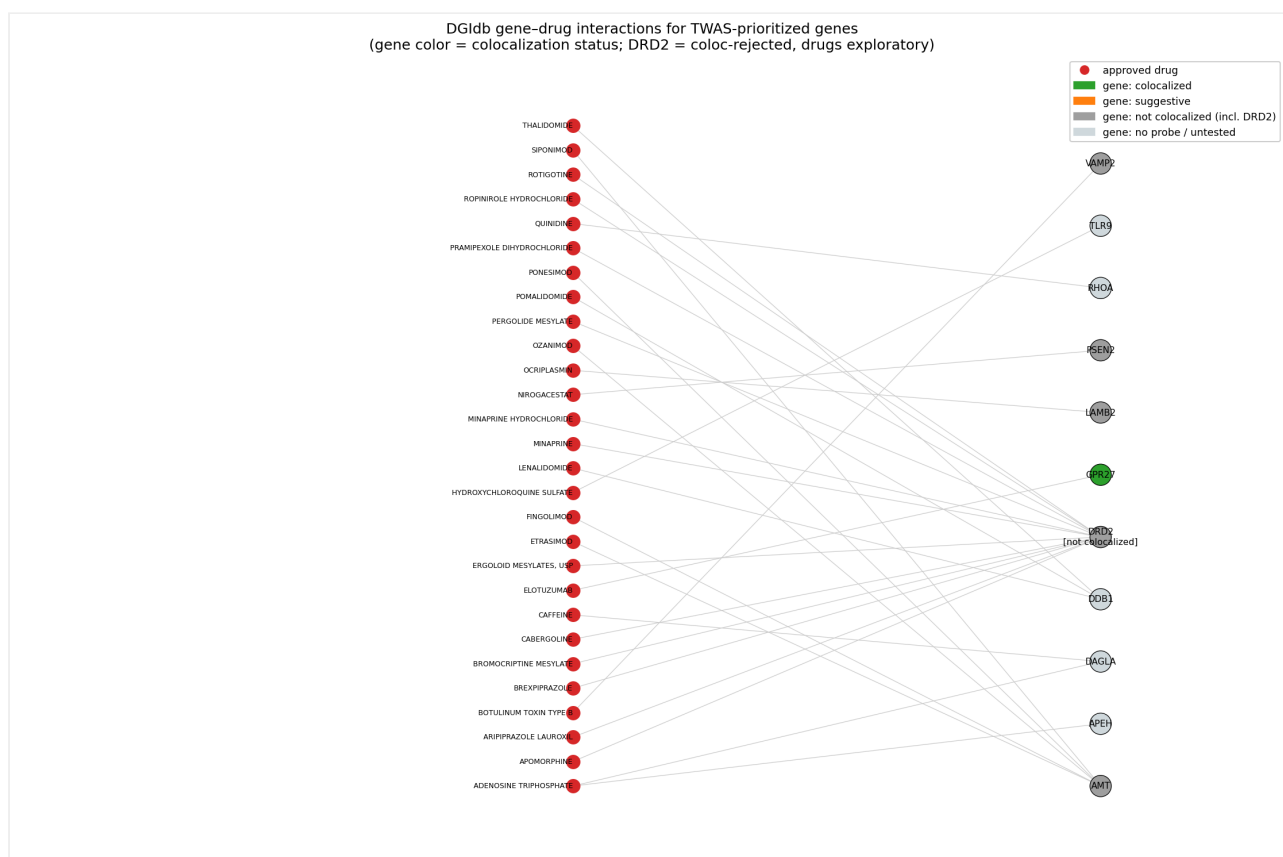


Figure 6. Direction-aware drug repurposing: (A) raw approved candidates (pre-confirmation); (B) colocalization-stratified prioritization, with the DRD2 dopaminergic cluster demoted to the exploratory tier.



Supplementary Figure S1. DGIdb gene-drug interactions for TWAS-prioritized genes (gene color = colocalization status; DRD2 = coloc-rejected, its drugs exploratory).